

| 《数据库系统原理》实验报告（3）                                                                                                                                                                                                                                                                                                                                                                                                                |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-----|----|----------|------------|--|--|----|----|----|-----|-------------|-------------|-------|-------------|--------|------|------------|--|------|----------|--|-------|-------------|--|
| 题目：数据库安全性                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 学号                                                                                                                                                                                                                                                                                                                                                                                                                              | 2452769     | 姓名          | 幸可函 | 日期 | 2026/4/7 |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 实验环境：                                                                                                                                                                                                                                                                                                                                                                                                                           |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 1. 操作系统：Windows 11                                                                                                                                                                                                                                                                                                                                                                                                              |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 2. 容器平台：Docker Desktop for Windows（版本：4.65.0）                                                                                                                                                                                                                                                                                                                                                                                   |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 3. 数据库镜像：oceanbase/oceanbase-ce:latest                                                                                                                                                                                                                                                                                                                                                                                          |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 4. 数据库系统：OceanBase Community Edition 4.3.5.5                                                                                                                                                                                                                                                                                                                                                                                    |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 实验步骤及结果截图：                                                                                                                                                                                                                                                                                                                                                                                                                      |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 1、建立数据库 Major，在 Major 数据库中创建表 Players                                                                                                                                                                                                                                                                                                                                                                                           |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| <table><tr><th colspan="3">表名：Players</th></tr><tr><th>字段</th><th>类型</th><th>约束</th></tr><tr><td>Sno</td><td>Varchar(10)</td><td>Primary Key</td></tr><tr><td>Sname</td><td>Varchar(20)</td><td>Unique</td></tr><tr><td>Ssex</td><td>Varchar(1)</td><td></td></tr><tr><td>Sage</td><td>SmallInt</td><td></td></tr><tr><td>Steam</td><td>Varchar(20)</td><td></td></tr></table>                                                  |             |             |     |    |          | 表名：Players |  |  | 字段 | 类型 | 约束 | Sno | Varchar(10) | Primary Key | Sname | Varchar(20) | Unique | Ssex | Varchar(1) |  | Sage | SmallInt |  | Steam | Varchar(20) |  |
| 表名：Players                                                                                                                                                                                                                                                                                                                                                                                                                      |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 字段                                                                                                                                                                                                                                                                                                                                                                                                                              | 类型          | 约束          |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| Sno                                                                                                                                                                                                                                                                                                                                                                                                                             | Varchar(10) | Primary Key |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| Sname                                                                                                                                                                                                                                                                                                                                                                                                                           | Varchar(20) | Unique      |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| Ssex                                                                                                                                                                                                                                                                                                                                                                                                                            | Varchar(1)  |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| Sage                                                                                                                                                                                                                                                                                                                                                                                                                            | SmallInt    |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| Steam                                                                                                                                                                                                                                                                                                                                                                                                                           | Varchar(20) |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| <pre>obclient(root@sys)[(none)]&gt; create database Major; Query OK, 1 row affected (0.098 sec)  obclient(root@sys)[(none)]&gt; use Major; Database changed  obclient(root@sys)[Major]&gt; create table Players( -&gt; Sno varchar(10), -&gt; Sname varchar(20), -&gt; Ssex varchar(1), -&gt; Sage smallint, -&gt; Steam varchar(20), -&gt; primary key(Sno), -&gt; unique(Sname)); Query OK, 0 rows affected (0.266 sec)</pre> |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| 2、向 Players 表中插入两个元组：                                                                                                                                                                                                                                                                                                                                                                                                           |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |
| ('200215121','NiKo','M',27,'G2')                                                                                                                                                                                                                                                                                                                                                                                                |             |             |     |    |          |            |  |  |    |    |    |     |             |             |       |             |        |      |            |  |      |          |  |       |             |  |

('200215122','mONESY','M',18,'G2')

```
obclient(root@sys)[Major]> insert into Players values('200215121','NiKo','M',27,'G2'),('200215122','mONESY','M',18,'G2')
Query OK, 2 rows affected (0.029 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

3、建立用户 Valve，授予用户 Valve 以系统特权，包括 create table、create user、alter user 和 drop user 等，并赋予其在授权的能力

```
obclient(root@sys)[Major]> create user Valve;
Query OK, 0 rows affected (0.166 sec)
```

```
obclient(root@sys)[Major]> grant all privileges on *.* to 'Valve' with grant option;
Query OK, 0 rows affected (0.098 sec)
```

4、删除 Valve 的 create user、alter user 和 drop user 的系统特权

```
obclient(root@sys)[Major]> revoke all privileges on *.* from 'Valve';
Query OK, 0 rows affected (0.067 sec)
```

5、在 Valve 用户下尝试查询 Players（注意使用 Major.Players）

```
obclient(root@sys)[Major]> alter user 'Valve' identified by '123456';
Query OK, 0 rows affected (0.129 sec)
```

```
sh-4.4# obclient -h127.1 -P2881 -uValve@sys -p
Enter password:
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221694471
Server version: OceanBase_CE 4.3.5.5 (r105000012025111711-c10174c0486c38f64a2222486986bbe15d5da0dc) (Built Nov 17 2025 12:20:08)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
obclient(Valve@sys)[(none)]> select * from Major.Players;
ERROR 1142 (42000): SELECT command denied to user 'Valve'@'%' for table 'Players'
[172.17.0.2:2882] [2026-04-09 06:24:18.913795] [YB42AC110002-00064F0089A077F3-0-0]
```

6、授予用户 Valve 对表 Players 的查询、插入、修改等对象特权，并赋予其再授权的能力

```
obclient(Valve@sys)[(none)]> exit;
Bye
sh-4.4# obclient -uroot@sys -h127.1 -P2881
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221706932
Server version: OceanBase_CE 4.3.5.5 (r105000012025111711-c10174c0486c38f64a2222486986bbe15d5da0dc) (Built Nov 17 2025 12:20:08)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

obclient(root@sys)[(none)]> grant select, insert, update on Major.Players to 'Valve' with grant option;
Query OK, 0 rows affected (0.080 sec)
```

7、在 Valve 用户下查询 Players（注意使用 Major.Players）

```
obclient(root@sys)[(none)]> exit;
\Bye
sh-4.4# obclient -h127.1 -P2881 -uValve@sys -p
Enter password:
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221708386
Server version: OceanBase_CE 4.3.5.5 (r105000012025111711-c10174c0486c38f64a2222486986bbe15d5da0dc) (Built Nov 17 2025 12:20:08)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

obclient(Valve@sys)[(none)]> select * from Major.Players;
+-----+-----+-----+-----+-----+
| Sno      | Sname | Ssex | Sage | Steam |
+-----+-----+-----+-----+-----+
| 200215121 | NiKo  | M    | 27   | G2    |
| 200215122 | mONESY | M    | 18   | G2    |
+-----+-----+-----+-----+-----+
2 rows in set (0.005 sec)
```

8、在 Valve 用户下再插入一个元组的数据('200215123','ZywOo','M',23,'Vit')，并 在 Valve 用户下查询（注意使用 Major.Players）

```
obclient(Valve@sys)[(none)]> insert into Major.Players values('200215123','ZywOo','M',23,'Vit');
Query OK, 1 row affected (0.012 sec)

obclient(Valve@sys)[(none)]> select * from Major.Players;
+-----+-----+-----+-----+-----+
| Sno      | Sname | Ssex | Sage | Steam |
+-----+-----+-----+-----+-----+
| 200215121 | NiKo  | M    | 27   | G2    |
| 200215122 | mONESY | M    | 18   | G2    |
| 200215123 | ZywOo | M    | 23   | Vit   |
+-----+-----+-----+-----+-----+
3 rows in set (0.003 sec)
```

9、删除 Valve 用户

```
obclient(Valve@sys)[(none)]> exit;
Bye
sh-4.4# obclient -uroot@sys -h127.1 -P2881
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221727174
Server version: OceanBase_CE 4.3.5.5 (r105000012025111711-c10174c0486c38f64a2222486986bbe15d5da0dc) (Built Nov 17 2025 12:20:08)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

obclient(root@sys)[(none)]> drop user 'Valve';
Query OK, 0 rows affected (0.091 sec)
```

出现的问题：

1.登录用户语句语法错误，误以为 Enter password:后会显示字符导致一直输入、登陆失败

```
sh-4.4# obclient -h 127.1 -u Valve@sys -p Major -P 2881
Enter password:
ERROR 1045 (42000): Access denied for user 'Valve'@'xxx.xxx.xxx.xxx' (using password: YES)
[0.0.0.0:0] [2026-04-09 06:19:50.675143] [Y0-00064F0081C077FC-0-0]
sh-4.4# obclient -h 127.1 -u Valve@sys -P 2881
ERROR 1045 (42000): Access denied for user 'Valve'@'xxx.xxx.xxx.xxx' (using password: NO)
[0.0.0.0:0] [2026-04-09 06:19:56.541356] [Y0-00064F0081C077FD-0-0]
sh-4.4# obclient -h 127.1 -u Valve@sys -P 2881;
Unknown suffix ' ' used for variable 'port' (value '2881')
obclient: Error while setting value '2881' to 'port'
```

2.试图在登录为 Valve 时给自己授权

解决方案：

1.在查阅文档后使用了正确的登录语句：

```
sh-4.4# obclient -h127.1 -P2881 -uValve@sys -p
Enter password:
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221694471
```

2.先退出登录、登录进 root 用户再给 Valve 授权